

ADMINISTRATION HVAC - CENTRAL PAU DUCT AND INSTRUMENTATION SCHEMATIC

OUTDOOR / ROOF PLANT

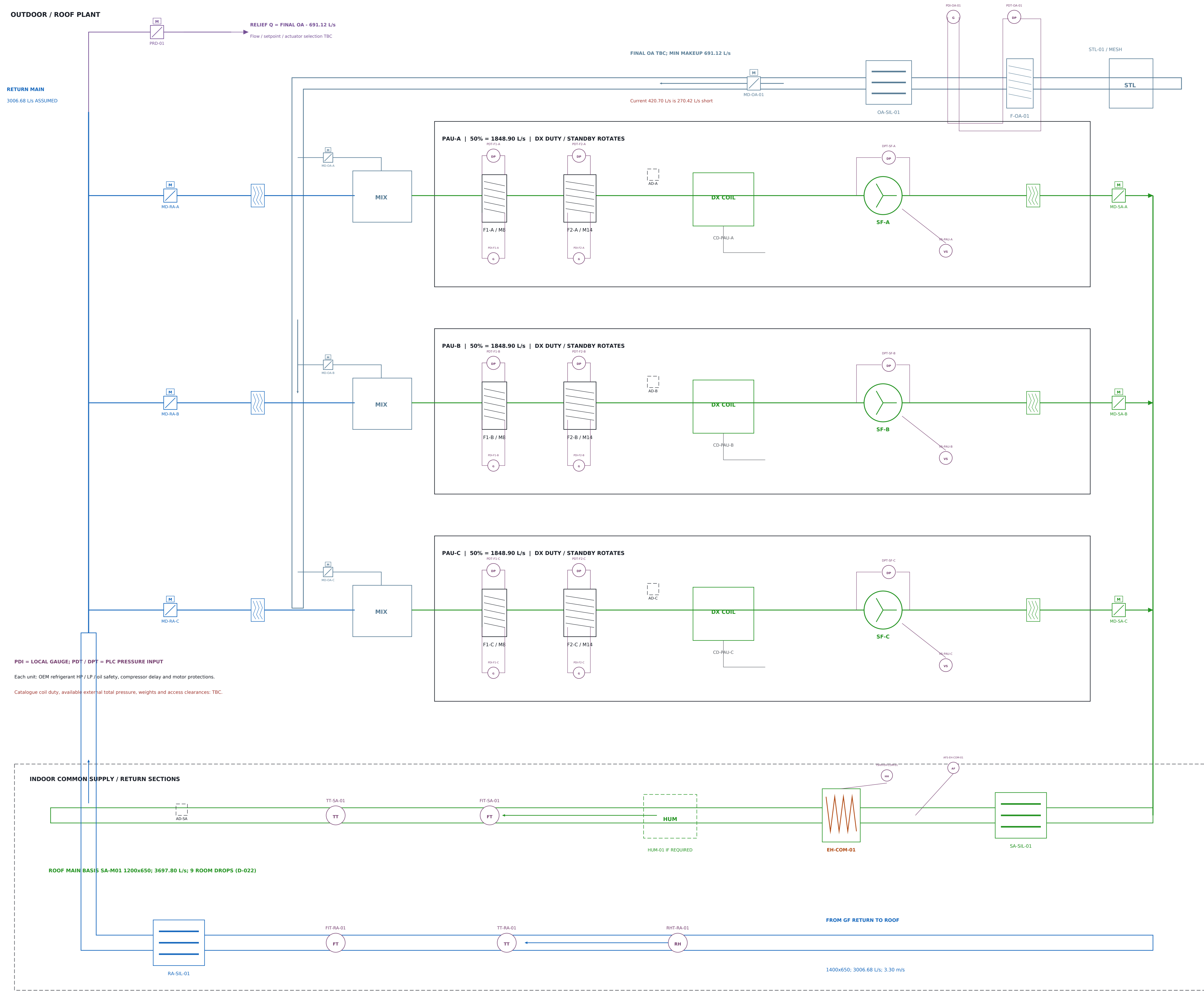
RETURN MAIN
3006.68 L/s ASSUMED

PDI = LOCAL GAUGE; PDT / DPT = PLC PRESSURE INPUT
Each unit: OEM refrigerant HP / LP / oil safety, compressor delay and motor protections.
Catalogue coil duty, available external total pressure, weights and access clearances: TBC.

INDOOR COMMON SUPPLY / RETURN SECTIONS

ROOF MAIN BASIS SA-M01 1200x650; 3697.80 L/s; 9 ROOM DROPS (D-022)

SOURCE: HAP / ADMIN ARCHITECTURE / C-H-SS-001 Rev0. All new equipment and instrument tags are proposed.



GENERAL NOTES / DESIGN HOLDS

1. All duct sizes are clear internal W x H in mm; round sizes are diameters. Add approved insulation outside these dimensions.
2. HAP reference supply 3697.80 L/s. Assumed return 3006.68 L/s. Final IAQ, outdoor air, revised thermal loads and OEM fan pressure remain open.
3. Room targets: +50 Pa normal rooms; +25 Pa kitchen, toilet and clean-agent relative to outdoors. Actual pressure / outward paths are unverified.
4. Confirmed fire stops the entire HVAC system. Fire ESD restart requires local manual reset; post-fire purge is a separate approved design.
5. Reference substation drawings guide drafting style only. Project requirements follow the supplied TAQA specification and accepted room basis.
6. 3 x 50% PAUs: two duty, one available. Any unit may be standby; programmed rotation and fault changeover apply.
7. Rev12 roof option: common heater / conditional humidifier in proposed weather-protected roof enclosure; room comfort heaters indoors. Roof suitability / final duty TBC.

FOR COORDINATION - NOT FOR CONSTRUCTION

Unconfirmed routes, levels, outdoor air and OEM selections: Insufficient data to verify.

CLIENT / DESIGN BASIS

TAQA TRANSMISSION

N-19082-RO-012 / Design Standardization for Pumping Stations

DRAWING TITLE

Administration HVAC - central PAU duct and instrumentation schematic

DRAWING NUMBER

ADM-HVAC-S-001

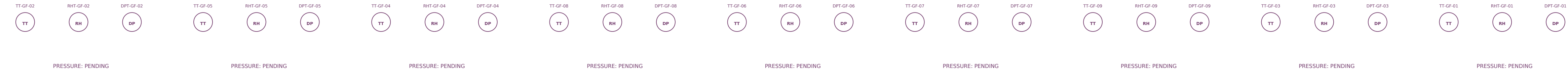
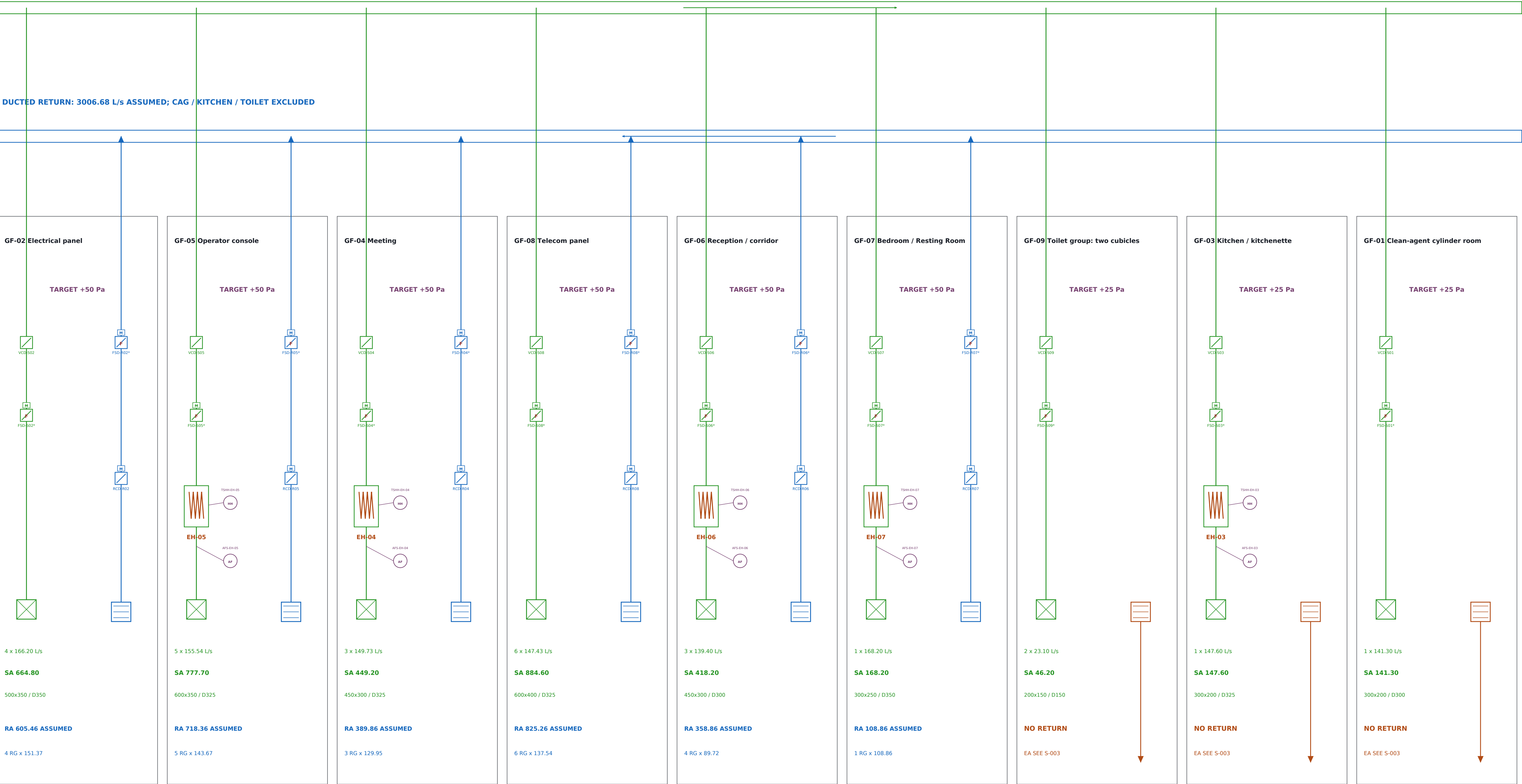
REV 11 | 07 Oct 2026 | 1/4

SCALE NTS | A1 / mm / L/s / Pa

Prepared: draft Checked / approved: pending

ADMINISTRATION HVAC - ROOM SUPPLY AND RETURN SCHEMATIC

PAU COMMON SUPPLY: 3697.80 L/s



DAMPERS / TERMINALS

Supply diffusers and return grilles: integral OBD, tagged neck, face, flow, plenum, test point and access. Drawn room branch paths have VCD; return pressure-control damper RCD is additional. Main and each room temperature / humidity / pressure values report to HPCP-01. Pressure sensor reference is outdoors, not the adjacent room.

ROOM AIRFLOW BALANCE

For the six +50 Pa return rooms, supply - return = 59.34 L/s as a NET outward allowance. It includes the net effect of transfer, relief and envelope outflow; it is not an additional HAP duct loss. Allocate actual paths before pressure verification. +25 Pa extract rooms must meet both minimum ACH and the full room / transfer balance.

SA = cooled mixed PAU air. Delivered room outdoor air must be checked separately from total supply / extract ACH.

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- HAP reference supply 3697.80 L/s. Assumed return 3006.68 L/s. Final IAQ, outdoor air, revised thermal loads and OEM fan pressure remain open.
- Room targets: +50 Pa normal rooms; +25 Pa kitchen, toilet and clean-agent relative to outdoors. Actual pressure / outward paths are unverified.
- Confirmed fire stops the entire HVAC system. Fire ESD restart requires local manual reset; post-fire purge is a separate approved design.
- Reference substation drawings guide drafting style only. Project requirements follow the supplied TAQA specification and accepted room basis.
- The common net equivalent area is 0.01 m² per +50 Pa return room. C=0.65, density=1.2 and delta-P=50 give 59.34 L/s offset per room.
- RCD return modulation is a proposed pressure-control provision. Confirm range, fail state and tuning; fixed flows are initial balancing values.
- FSD shown with * is conditional on approved fire compartmentation. Exact ratings, locations and fire positions remain TBC.

FOR COORDINATION - NOT FOR CONSTRUCTION

Unconfirmed routes, levels, outdoor air and OEM selections: Insufficient data to verify.

CLIENT / DESIGN BASIS

TAQA TRANSMISSION

N-19082-RO-012 / Design Standardization for Pumping Stations

DRAWING TITLE

Administration HVAC - room supply and return schematic

DRAWING NUMBER

ADM-HVAC-S-002

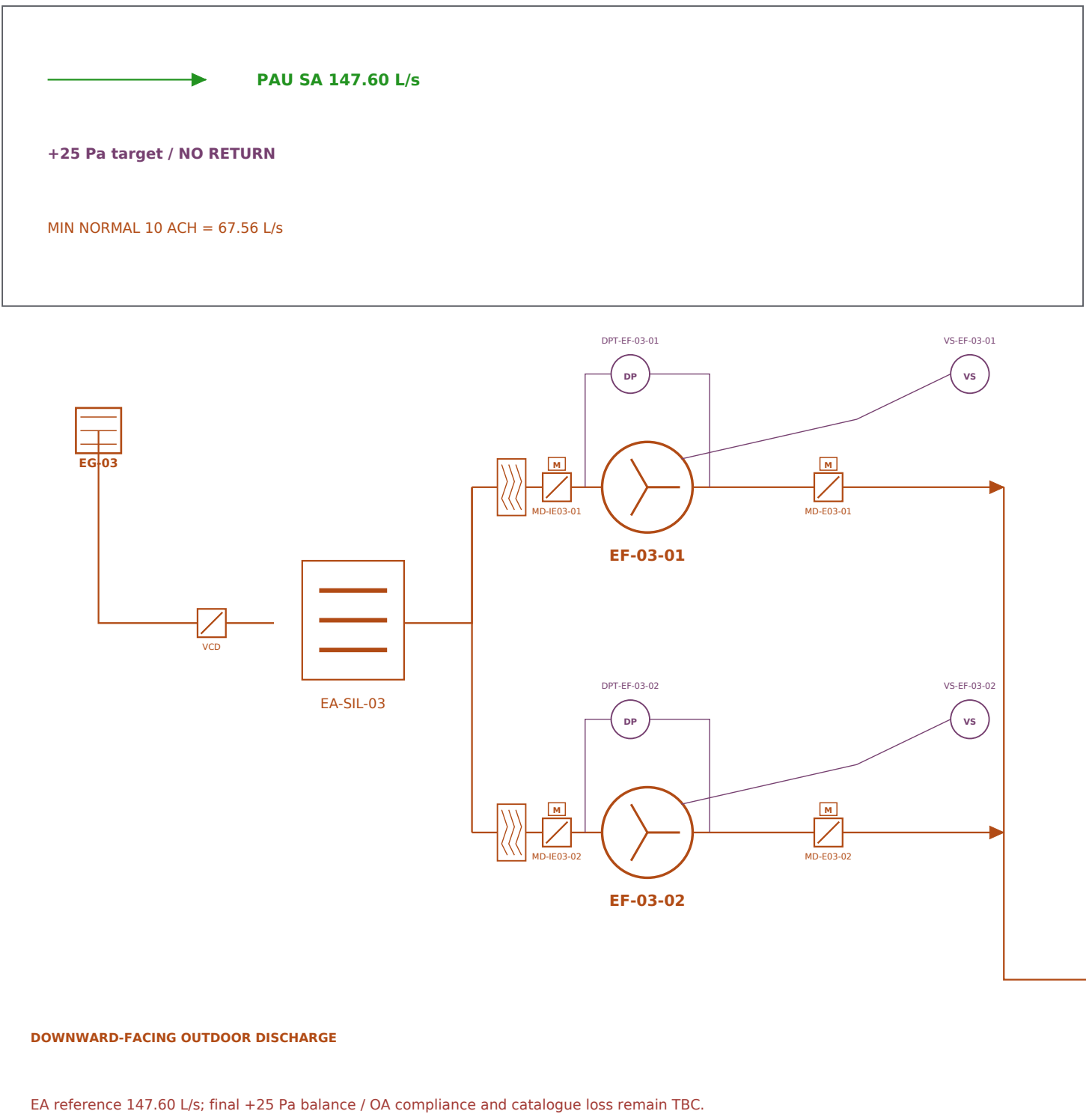
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SCALE NTS | A1 / mm / L/s / Pa

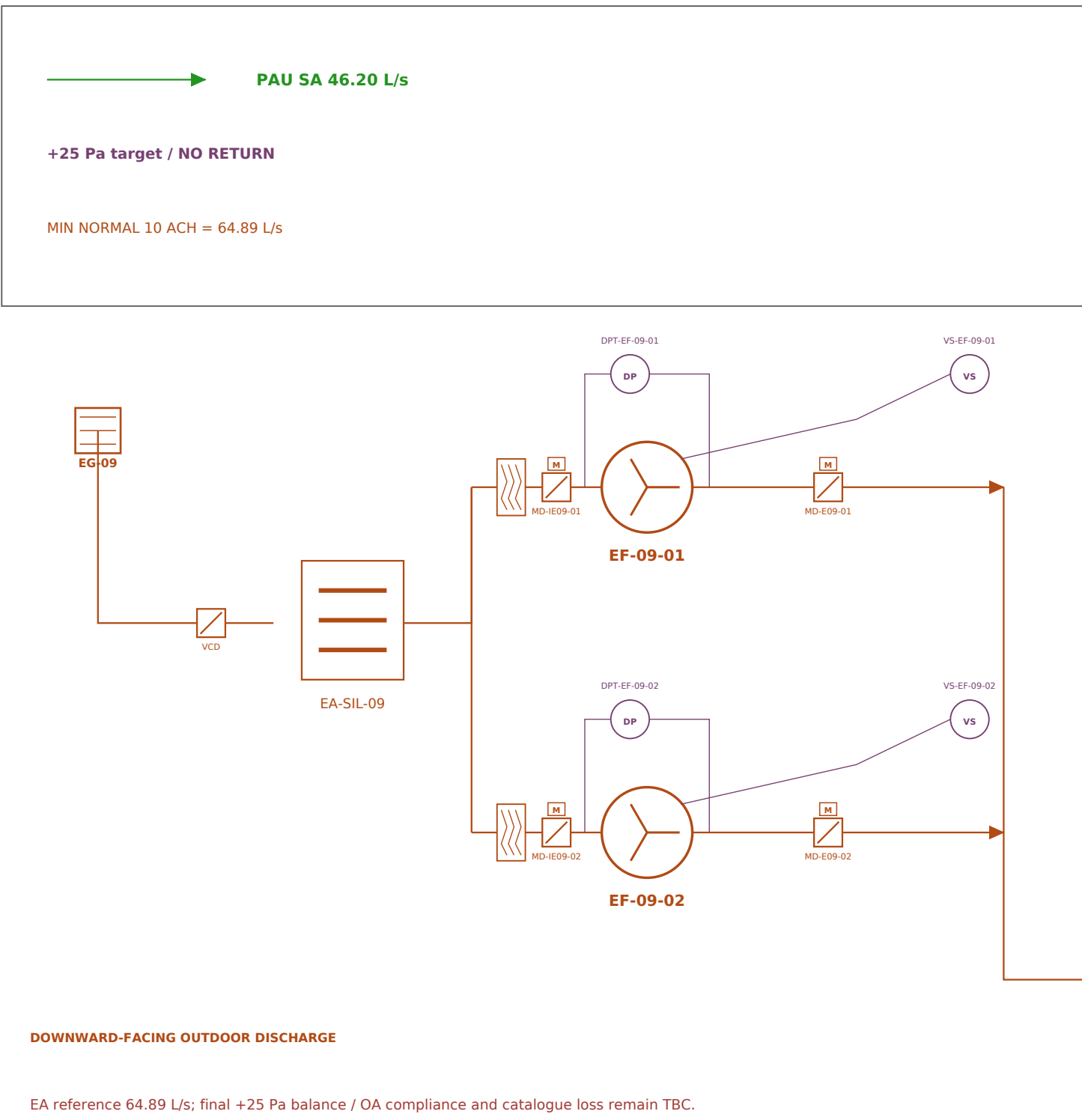
Prepared: draft / Checked / approved: pending

ADMINISTRATION HVAC - DEDICATED EXTRACTS AND CABLE BASEMENT

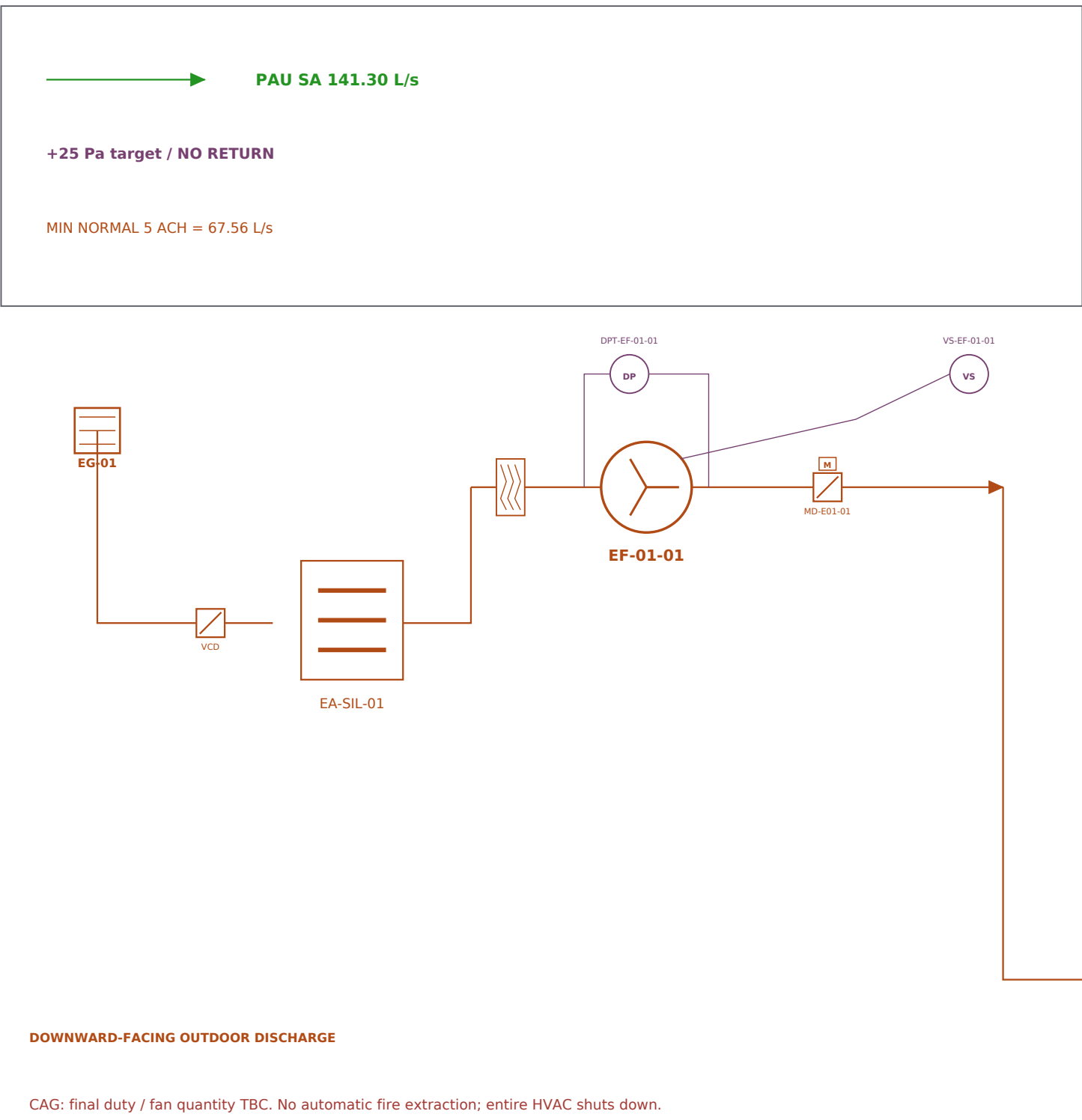
GF-03 KITCHEN / KITCHENETTE



GF-09 TOILET GROUP: TWO CUBICLES

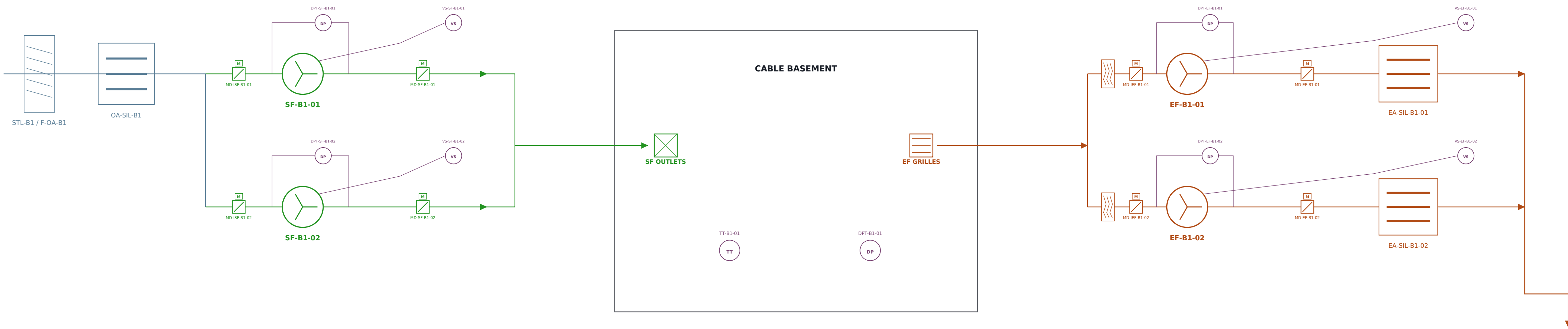


GF-01 CLEAN-AGENT CYLINDER ROOM



CABLE BASEMENT - SEPARATE VENTILATION CIRCUIT

AIRFLOW DUTY TBC: $Q \geq \max(4 \times V / 3.6, 1000 \times \text{heat_kW} / (1.06 \times 1.0216 \times 5))$ L/s



NORMAL / FIRE / POST-FIRE

Normal fans satisfy temperature, fresh-air, ACH and pressure criteria together. Confirmed fire stops ALL HVAC fans and heaters, including CAG and basement. Fire damper positions follow the approved compartment and cause-and-effect design. Same-fan post-fire purge remains a separate proposal; no smoke-control fan rating or automatic purge is established here.

Basement plan, duct sizes, equipment clearances, penetrations, fan curve and full pressure loss circuit: TBC.

GENERAL NOTES / DESIGN HOLDS

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- Room targets: +50 Pa normal rooms; +25 Pa kitchen, toilet and clean-agent relative to outdoors. Actual pressure / outward paths are unverified.
- Confirmed fire stops the entire HVAC system. Fire ESD restart requires local manual reset; post-fire purge is a separate approved design.
- Reference substation drawings guide drafting style only. Project requirements follow the supplied TAQA specification and accepted room basis.
- Kitchen and toilet: 2 x 100% extract with automatic duty rotation. No central return from these rooms.
- Clean-agent: normal 5 ACH minimum; dedicated extract, fan quantity / pickup level / agent suitability TBC.
- Basement: separate 2 x 100% supply + 2 x 100% extract; greater of 4 ACH and heat removal. Volume and heat load are missing.

FOR COORDINATION - NOT FOR CONSTRUCTION

Unconfirmed routes, levels, outdoor air and OEM selections: Insufficient data to verify.

CLIENT / DESIGN BASIS

TAQA TRANSMISSION

N-19082-RO-012 / Design Standardization for Pumping Stations

DRAWING TITLE

Administration HVAC - dedicated extracts and cable basement

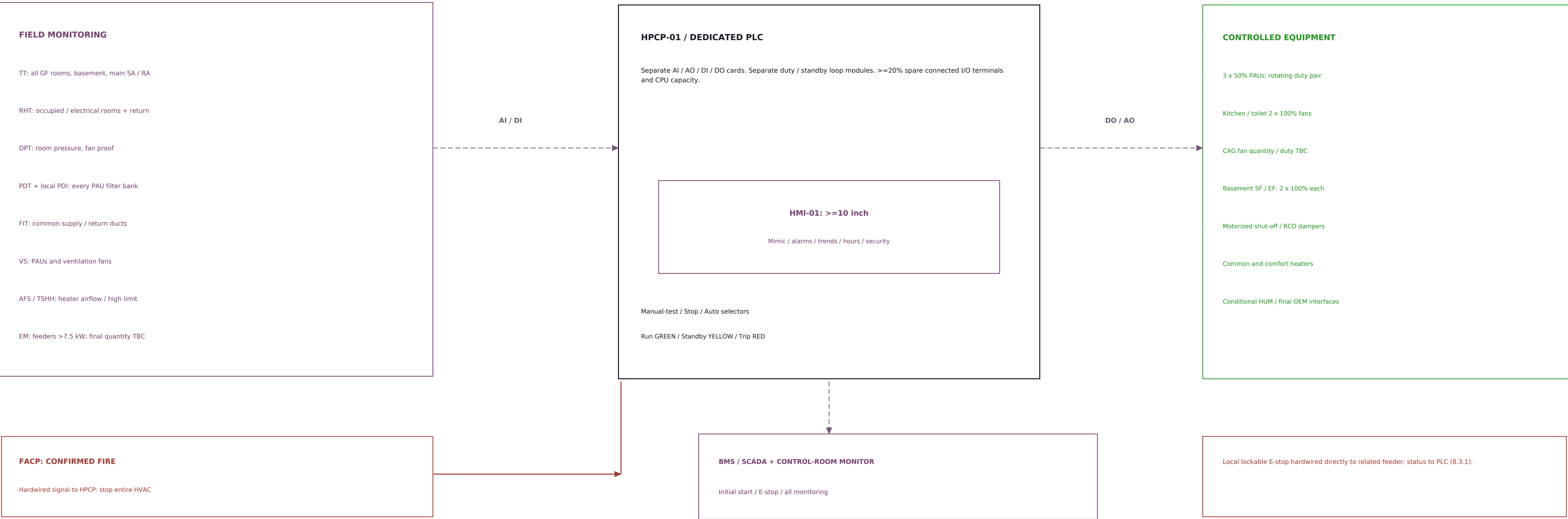
DRAWING NUMBER

ADM-HVAC-S-003

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SCALE NTS | A1 / mm / L/s / Pa

Prepared: draft Checked / approved: pending



PRELIMINARY WIRED I/O ALLOCATION

Type	Shown points	20% spare minimum	Minimum channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

Full unique tag register, equipment list and individual points are included as CSV files in the CAD package. COMM values and OEM-internal safety circuits are not counted as wired AI/DI points. All new tags are draft allocations.

DRAFT OPERATING SEQUENCE / CAUSE-AND-EFFECT

Trigger	Input	Equipment action	Damper / permissive action	Restart / limits
Confirmed fire	FACP -> HPCP	STOP all PAUs, extract and basement fans; heaters off	Damper fire positions per approved cause-and-effect; not inferred	Local manual reset after fire clears
Normal start	HMI / BMS start + Auto	Prove duty path dampers; start chosen duty pair and duty extract fans	Standby PAU and parallel fan inlet / outlet paths isolated	Sequence / delays by approved OEM logic
Duty PAU fault	Unit fault / fan DP	Alarm; isolate failed unit; enable available alternate	Maintain two healthy 50% units when available	Automatic fault changeover; minimum-runtime delay TBC
Low / high room pressure	Room DPT	Proposed RCD / OA or extract-speed adjustment within airflow constraints	Respect cooling, fresh-air and minimum ACH limits	Alarm if targets cannot be met; revise supply / balance
Extract duty fan fault	Fan fault / proof	Start standby for kitchen / toilet / basement	Prove available path; isolate failed path	CAG standby arrangement TBC
Heater high limit / no flow	TSHH / AFS	Heater OFF	Fans / dampers per OEM sequence; fire overrides	Manual or OEM reset only after fault clears
Power loss / return	Mains / feeder status	HVAC stops; normal restart on restored power	Fire ESD latch overrides automatic restart	Automatic power-return restart only without fire latch
Post-fire purge	Separate approved command	Not enabled by this drawing	Agent, fire fan suitability and makeup to be approved	No automatic purge assumed

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3. Room targets: +50 Pa normal rooms; +25 Pa kitchen, toilet and clean-agent relative to outdoors. Actual pressure / outward paths are unverified.
4. Confirmed fire stops the entire HVAC system. Fire ESD restart requires local manual reset; post-fire purge is a separate approved design.
5. Reference substation drawings guide drafting style only. Project requirements follow the supplied TAQA specification and accepted room basis.
6. Control logic, alarm limits, delays, communication protocol and actuator fail positions require I&C / OEM approval.
7. Room DPTs beyond the TAQA minimum and extract fan pressure proof are explicit added provisions, not quoted minimum requirements.
8. Point schedule is preliminary. CAG quantity, conditional fire dampers, humidifier and final energy-meter feeder count can change I/O.

FOR COORDINATION - NOT FOR CONSTRUCTION

Unconfirmed routes, levels, outdoor air and OEM selections: Insufficient data to verify.

CLIENT / DESIGN BASIS

TAQA TRANSMISSION

N-19082-RO-012 / Design Standardization for Pumping Stations

DRAWING TITLE

Administration HVAC - controls, I/O and cause-and-effect

DRAWING NUMBER

ADM-HVAC-S-004

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SCALE NTS | A1 / mm / L/s / Pa

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